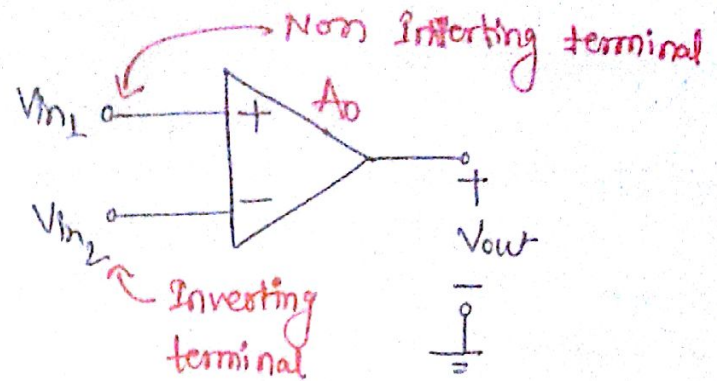


Lecture - 42

* The op-amp is an amplifier which has two inputs and one output.

* If we give V_{in1} input and take zero input at V_{in2} then we get output which has positive gain.



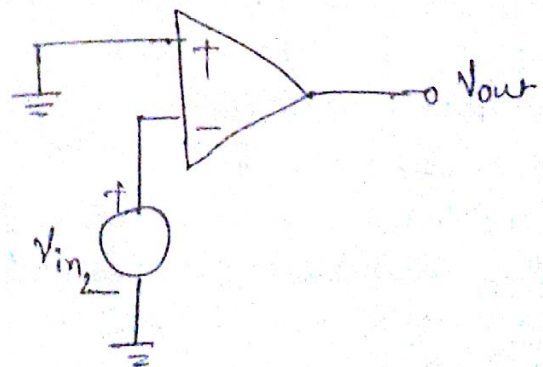
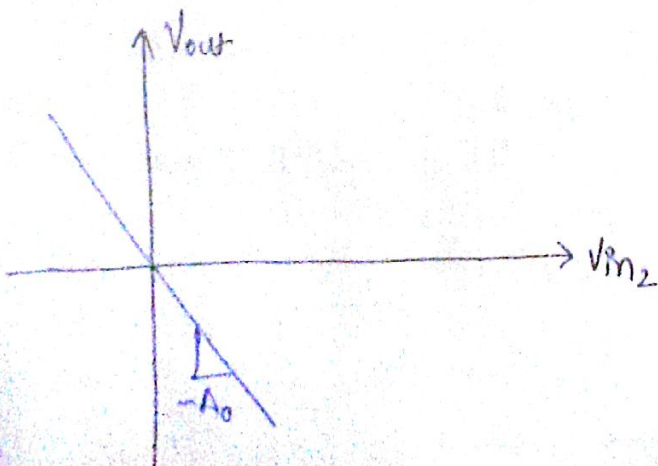
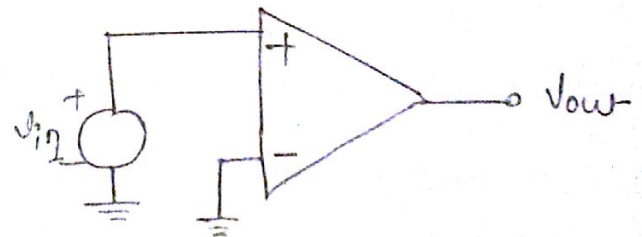
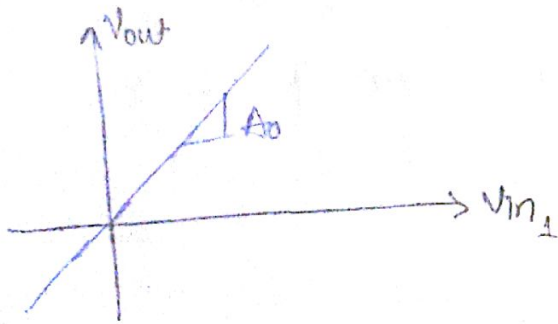
* If we give V_{in2} input and take zero input at V_{in1} then we get output which has negative gain.

$$V_{out} = A_0 V_{in1} \quad \leftarrow V_{in2} \text{ is zero}$$

$$V_{out} = -A_0 V_{in2} \quad \leftarrow V_{in1} \text{ is zero}$$

$$V_{out} = A_0 (V_{in1} - V_{in2}) \quad \leftarrow \text{Both signal are applied}$$

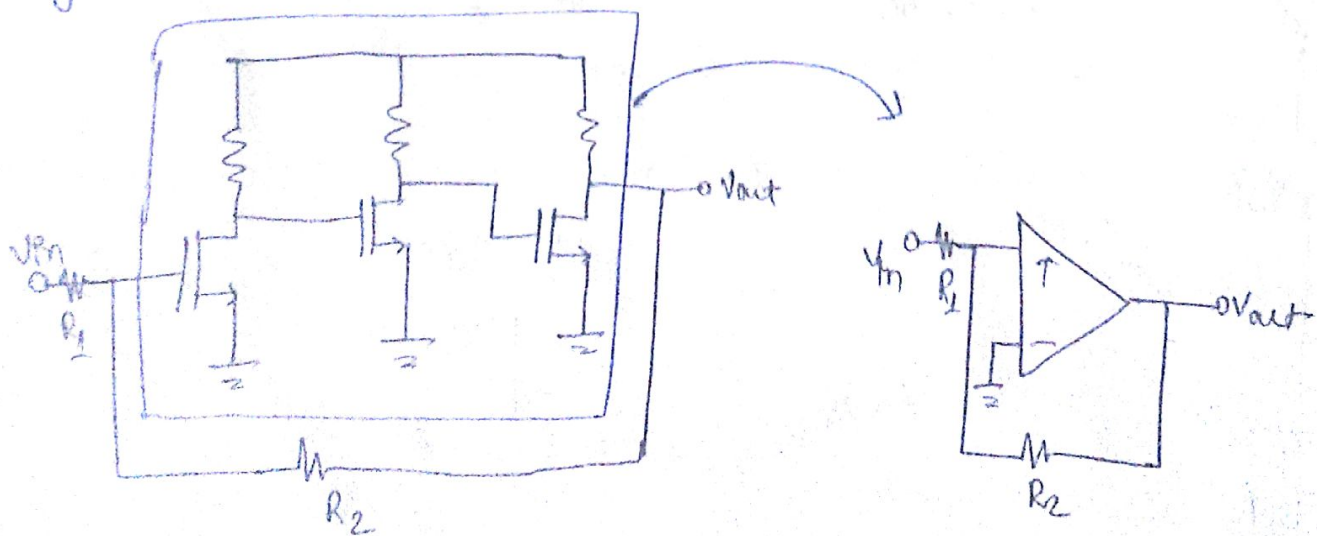
* Input/output characteristic:-



	Ideal op-Amp	741	Modern Integrated opamp
A_o	∞	100,000	50-100
Input Impedance	∞	Several Megaohms	Several Megaohms
Output Impedance	0	$< 100 \Omega$	Several kilohms
Speed	$BW = \infty$	1.5 MHz	Several MegaHz
Power dissipation	0	80 mW	5-10 mW
Cost	0	$< \$1$	$\$ 10^{-5}$

Observations:-

- ① If V_{out} is a few volts then $A_o \approx 1000$ and $V_{in1} - V_{in2} \approx$ a few mV.
- ② If we can visualize a complex circuit as an op-amp in the analysis becomes simpler.



We just have to view thing as an op-amp.

* Non-Inverting Amplifier:-

* We feed this divided voltage to the inverting terminal. We feed input to the non-inverting amplifier.

Case I $A_o \rightarrow \infty$

$$V_{out} = \underbrace{(V_{in_1} - V_{in_2})}_{\text{very small}} \underbrace{A_o}_{\text{very large}}$$

finite

$$\Rightarrow V_{in_1} \approx V_{in_2}$$

If the gain of op-amp is very high then $(V_{in_1} - V_{in_2})$ must be very small which means V_{in_1} and V_{in_2} must be close to each other.

$$V_{out} = \left(\frac{V_{in}}{R_2} \right) (R_1 + R_2)$$

We can write this because we are assuming no current flowing through the op-amp, all the current is flowing through R_1 and R_2 .

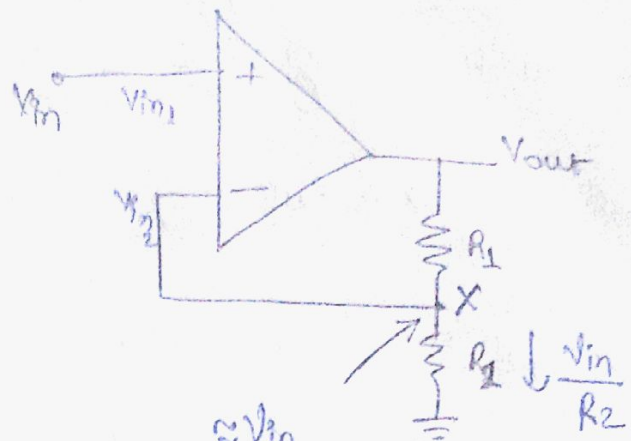
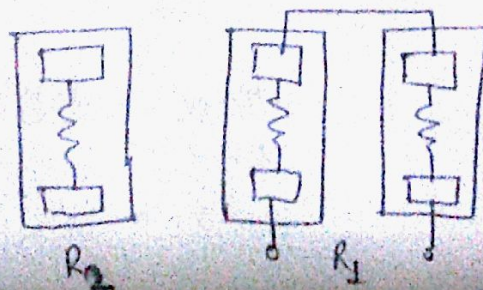
Therefore

$$\frac{V_{out}}{V_{in}} = \left(1 + \frac{R_1}{R_2} \right)$$

Voltage Gain of non-inverting Amp

→ It does not depend on the A_o and temperature because R_1 and R_2 always be constant.

for getting gain = 3 then we just have to get R_2 half of R_1 .



$\approx V_{in}$
because
 $V_{in_1} \approx V_{in_2}$

Case II A_0 not very high

Since A_0 is not very large so we can not assume $V_{in1} \approx V_{in2}$

$$V_x = V_{out} \frac{R_2}{R_1 + R_2}$$

$$V_{out} = (V_{in1} - V_{in2}) A_0$$

$$\Rightarrow (V_{in} - V_x) A_0 = V_{out}$$

$$\Rightarrow \left(V_{in} - \frac{V_{out} R_2}{R_1 + R_2} \right) A_0 = V_{out}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{A_0}{1 + \left(\frac{R_2}{R_1 + R_2} \right) A_0}$$

Open loop gain $\rightarrow A_0$

Closed loop gain $\rightarrow \frac{V_{out}}{V_{in}}$

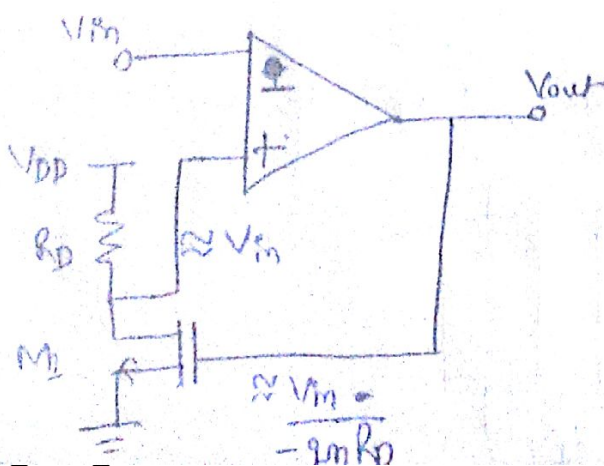
This equation tells us that for previous equation, we don't need A_0 to be ∞ , all we need is $\frac{R_2}{R_1 + R_2} A_0$ much greater than 1.

If $\frac{R_2}{R_1 + R_2} A_0 \gg 1$ then $\frac{V_{out}}{V_{in}}$ relatively independent of A_0 .

* We call A_0 as open loop gain means its gain is the gain which has op-amp without any connection.

* We call $\frac{V_{out}}{V_{in}}$ as closed loop gain means it is the gain which we get after connection of op-amp with some circuit.

Quiz:



Assume A_0 is very high then what is $\frac{V_{out}}{V_{in}}$?

Since voltage gain for CS stage $= -g_m R_D$

Ans

$$V_{out} = -\frac{V_{in}}{g_m R_D}$$

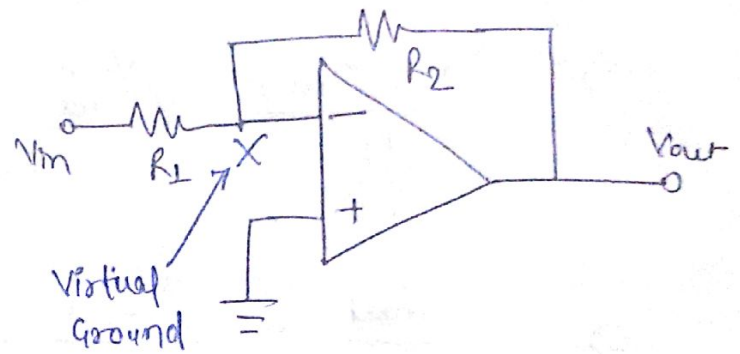
$$\Rightarrow \boxed{\frac{V_{out}}{V_{in}} = -\frac{1}{g_m R_D}}$$

We have connected it to non inverting terminal because we want to this whole loop to contain one inversion so it should be on non inverting terminal.

* Inverting Amplifier:-

Case 1 $A_o \rightarrow \infty$

Since V_{in1} is 0 so V_{in2} is closer to zero $V_{in2} = V_x$ is virtual ground.



$$\text{Current through } R_1 = \frac{V_{in} - V_x}{R_1} \approx \frac{V_{in}}{R_1} \quad [V_x \approx 0]$$

Since current cannot go through op-amp because it has high output impedance.

$$\frac{V_{in}}{R_1} R_2 = V_x - V_{out}$$

$$\boxed{\frac{V_{out}}{V_{in}} \approx -\frac{R_2}{R_1}}$$

It is high precise value for the closed loop gain.

* Input impedance $\approx R_1$.

Case II A_o not very large

$$V_{out} = (V_{in1} - V_{in2}) A_o$$

$$\Rightarrow V_x = \frac{V_{out}}{-A_o} \quad \text{by } V_{in1} = 0$$

$$\frac{V_{in} - V_x}{R_1} = \frac{V_x - V_{out}}{R_2}$$

$$\Rightarrow \frac{V_{in} + \frac{V_{out}}{A_o}}{R_1} = \frac{-\frac{V_{out}}{A_o} - V_{out}}{R_2}$$

$$\Rightarrow \boxed{\frac{V_{out}}{V_{in}} = \frac{-1}{\frac{R_1}{R_2} + \frac{1}{A_o} \left(\frac{R_1 + R_2}{R_1} \right)}}$$

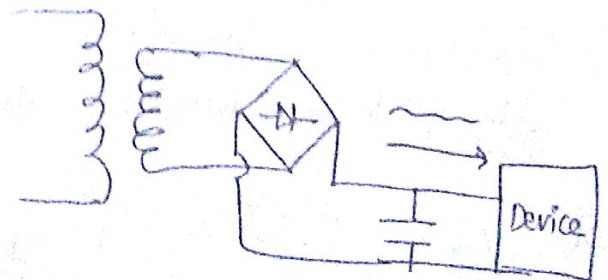
Closed loop gain
of ~~non~~ Inverting
Amp.

If $\frac{1}{A_o} \left(\frac{R_1 + R_2}{R_1} \right) \ll \frac{R_1}{R_2}$ then we get the closed loop gain,
which is relatively independent of A_o .

Lecture - 4.3

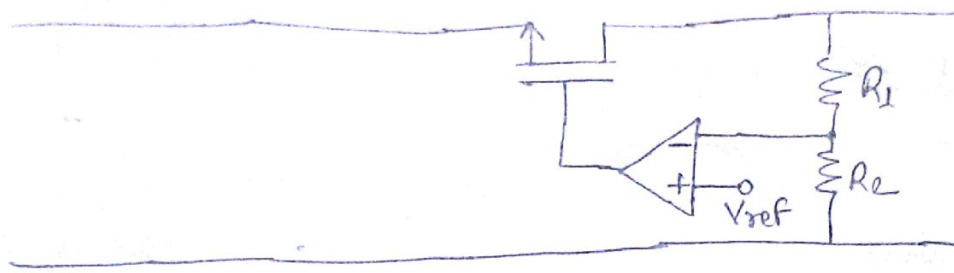
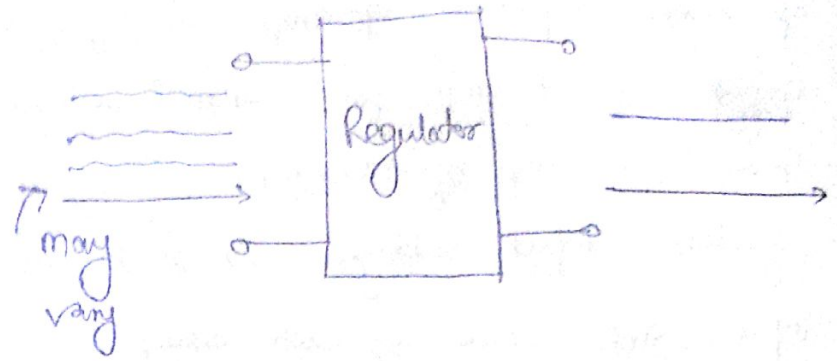
* Example of application!:- Voltage regulator

→ In this, input may have some fluctuations which may result in fluctuations in output which is not appropriate for the device.



So we add something between them a circuitry called a voltage regulator.

* The job of the regulator is to take voltage that might vary and produce a constant voltage (regulated voltage).



* We have input and output and if input voltage is wanted to change but output does not want to change then we have to take a mechanism to control input voltage. We use a MOSFET as a variable resistor.

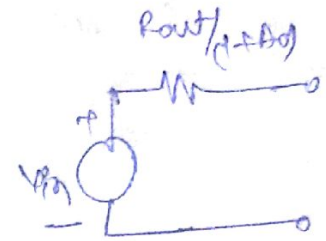
* To vary this voltage we can use gate voltage, gate voltage goes down or goes up the resistance changes. We have to measure output voltage and determine if it is a right value. This output voltage may be high so we have to bring it down little bit, attenuated by some factor so we use R_1 and R_2 . We just measure voltage at combination point of R_1 and R_2 . That voltage is related by some factor to the output voltage.

* We have to tell MOSFET to adjust its resistance and hence output voltage depending upon the voltage we are measuring. After comparing with a reference voltage V_{ref} by using an op-amp.

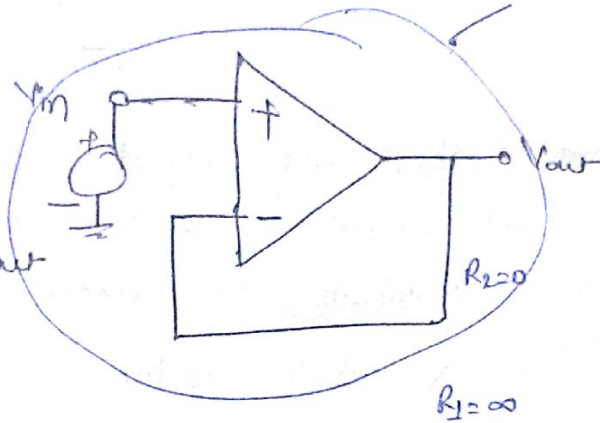
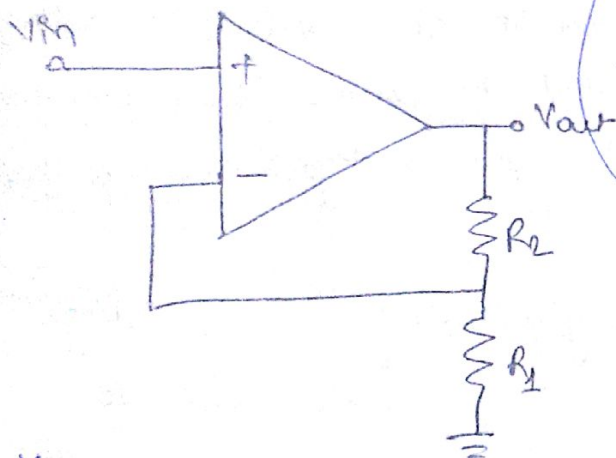
The output of op-amp is fed to the gate..

* If the input voltage fluctuates up then output voltage also goes up then input of op-amp also goes up and goes above V_{ref} then output of op-amp goes down because ~~the~~ ~~input~~ inverting terminal. Then resistance of MOSFET increases. Therefore this loop tries to regulate input voltage so that after everything settle both inputs are close to each other.

$$V_{out} \approx V_{ref} \left(1 + \frac{R_1}{R_2} \right)$$



* Unity gain Buffer:-



$$\frac{V_{out}}{V_{in}} \approx 1$$

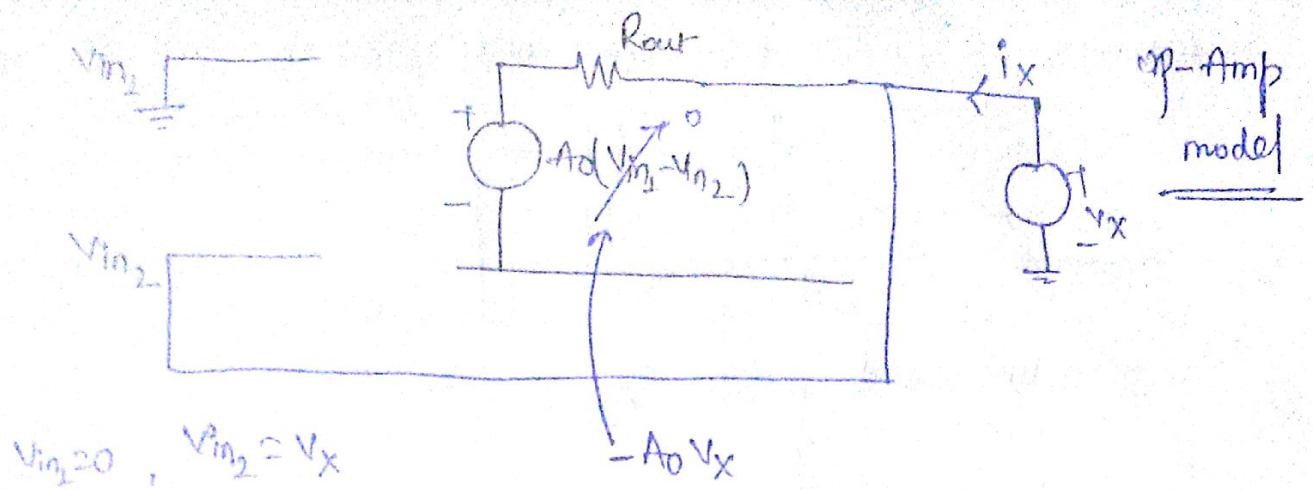
$$\frac{V_{out}}{V_{in}} \approx 1 + \frac{R_2}{R_1}$$

As a special case if $\frac{R_2}{R_1}$ is very small i.e. R_2 is 0 & R_1 is ∞ then we have unity gain buffer.

* Input impedance = ∞ so that means it can sense voltages without loading the circuit.

$$V_{Ther} = V_{in}$$



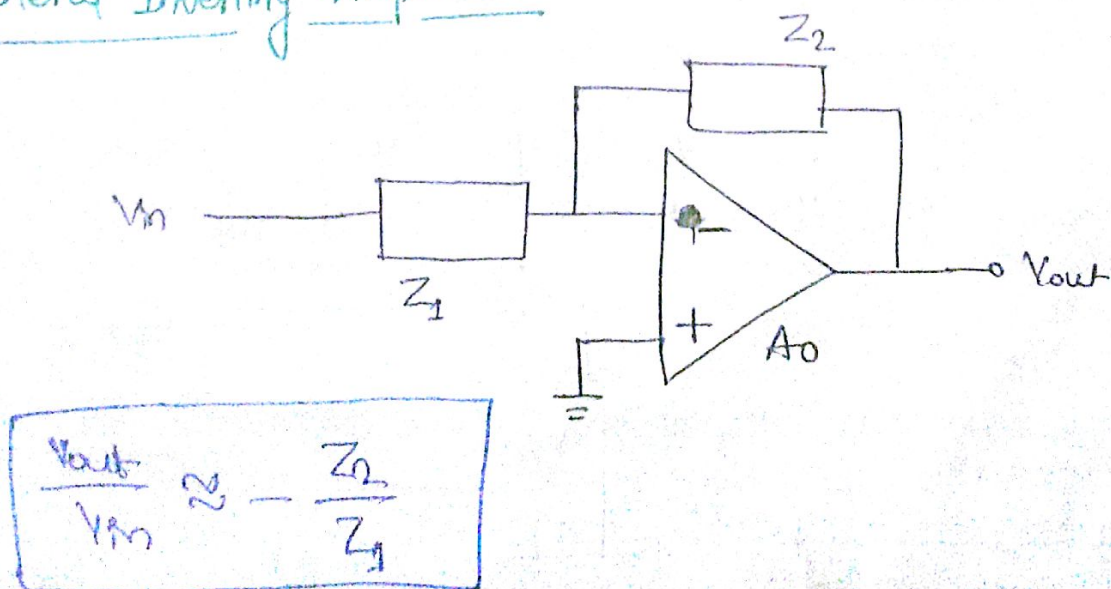


$$i_x = \frac{V_x + A_o V_x}{R_{out}} \Rightarrow \boxed{\frac{V_x}{i_x} = \frac{R_{out}}{1 + A_o}}$$

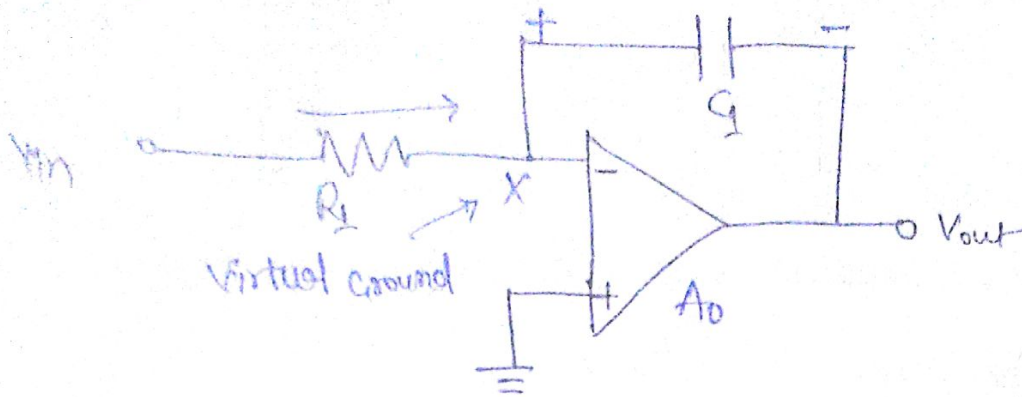
This says that even though when I bought a 741 just by itself, it comes with output resistance of 75Ω after I placed it in this configuration namely shorting output and ~~the~~ input the resistance that now has this presence from the outside world is as low as 75Ω divided by 100000 . The resistance of ~~the~~ unity gain buffer is extremely small depending R_{out} & A_o .

* This buffer is very good driver, it can drive ~~a~~ heavy load because its output resistance is very low.

General Inverting Amplifier:-



* Integrator In this we use $Z_1 = R_1$ and $Z_2 = C_1$.



If A_o is large,

$$\frac{V_{out}}{V_{in}} \approx - \frac{1/C_1}{R_1}$$

Pole at origin

$$\boxed{\frac{V_{out}}{V_{in}} \approx - \frac{1}{R_1 C_1 s}}$$

$$\left[\frac{1}{s} \leftrightarrow \int dt \right]$$

In time domain,

$$\text{Current through } R_1 \approx \frac{V_{in}}{R_1}$$

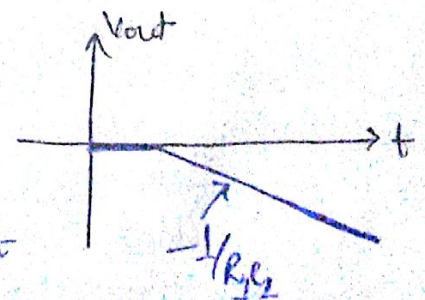
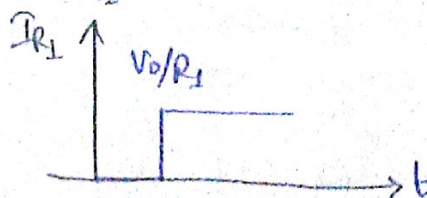
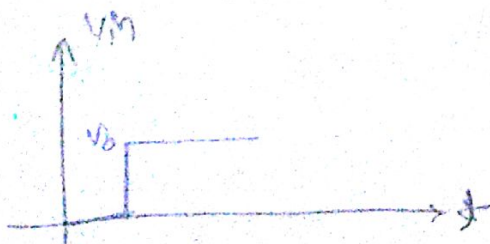
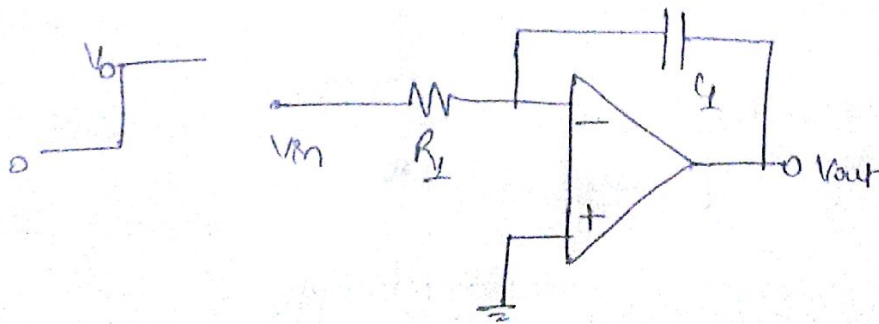
$$\begin{aligned} \text{Current through a cap.} \\ &= C \frac{dv}{dt} \end{aligned}$$

$$- \frac{dV_{out}}{dt} C_1 = \frac{V_{in}}{R_1}$$

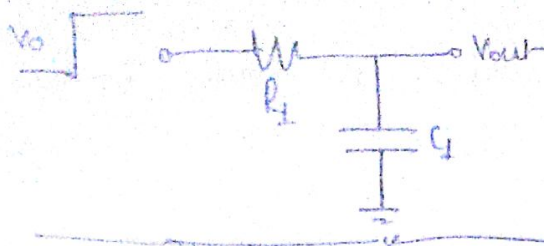
$\Rightarrow$

$$\boxed{V_{out} = - \frac{1}{R_1 C_1} \int V_{in} \cdot dt}$$

Example:-



File note.



In this current through R_1 is not equal to $\frac{V_{in}}{R_1}$

If A_o is ~~not~~ large,

$$V_{out} = A_o (V_{in1} - V_{in2})$$

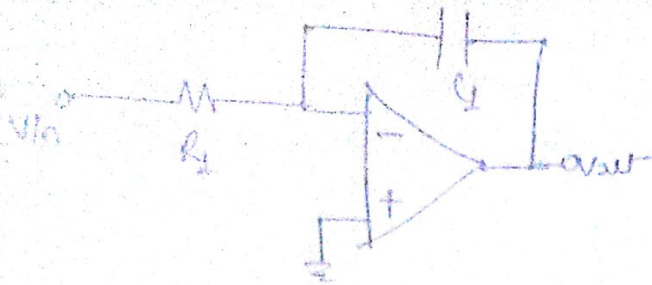
$$V_x = -\frac{V_{out}}{A_o}$$

$$\frac{V_{in} - V_x}{R_1} = \frac{V_x - V_{out}}{1/C_1 s}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{-1}{\frac{1}{A_o} + \left(1 + \frac{1}{A_o}\right) R_1 C_1 s}$$

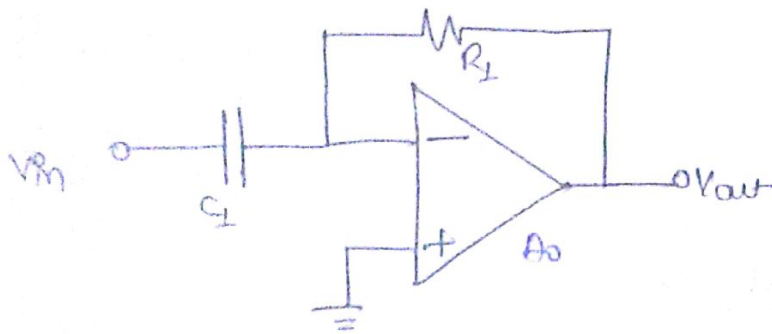
← Pole not at origin

Since in previous case, we have pole at origin and in this case, we have pole not at origin. Now, in previous case, if frequency of operation becomes smaller and smaller, we see that gain becomes larger and larger. So if s goes to very small value the gain of the ckt goes to very large value. It is unbounded so long as A_o is very large. On the other hand, if s becomes smaller and smaller then eventually we end up with only A_o so the gain is not unbounded it has a limit.



does not work in the lab.

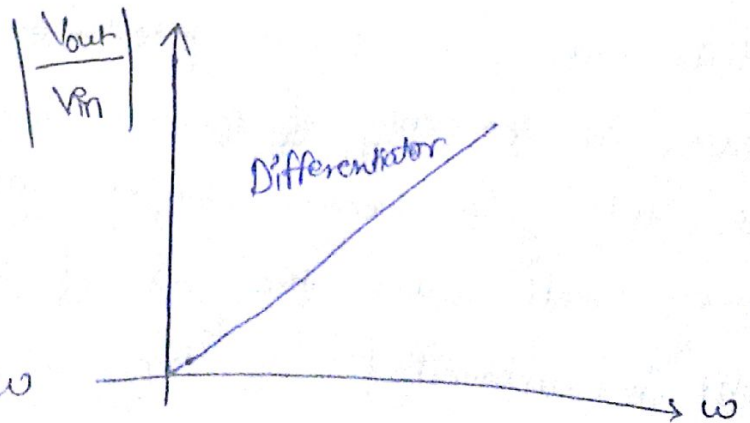
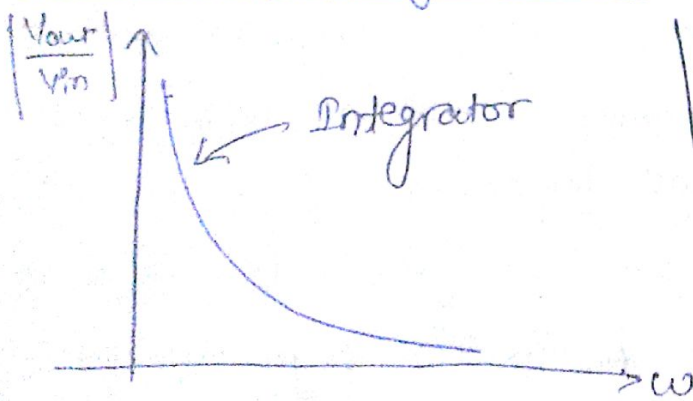
* Differentiator :- We take $Z_1 = C_1$ and $Z_2 = R_1$



If A_o is large,

$$\frac{V_{out}}{V_{in}} \approx - \frac{R_1}{1/C_1 s} \approx - R_1 C_1 s$$

Plot the frequency response:

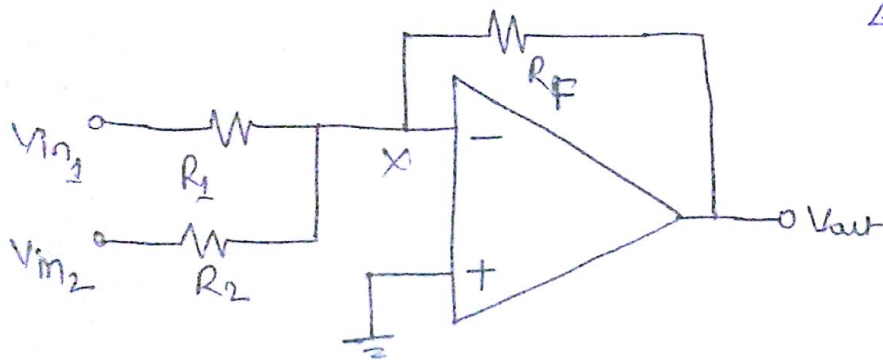


→ An Integrator amplifies low frequency and attenuates high frequency.

→ A differentiator amplifies high frequencies and attenuates low frequency.

* A differentiator does not use much because it amplifies high frequency noise also.

* Voltage Adder (Summer) :-



← This topology capable of converting V_{in} to a current and lets this current flow from R_f .

$$\text{Total current through } R_f = \frac{V_{in1}}{R_1} + \frac{V_{in2}}{R_2}$$

$$\Rightarrow V_{out} = -R_f \left(\frac{V_{in1}}{R_1} + \frac{V_{in2}}{R_2} \right)$$

→ It says that the output voltage is summation of two input voltages and depending on values of R_1 and R_2 , we can give different weighing factor to these two.